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SWEDISH ORDOVICIAN LIMESTONES: PROBLEMS IN CLARIFYING THEIR DIRECTIONS OF MAGNETIZATIONS

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The easterly region of the Swedish Ordovician limestones (the Orthoceras Limestone) shows a fairly stable bulk magnetization of high positive (downward) inclination. A closer scrutiny reveals, however, that this magnetization is of polyphase structure, involving at least two axes of magnetization. The components have similar stability spectra, and alternating field demagnetization cannot separate them. Thermal demagnetization is more effective but no stable end-point has been reached.

1. Introduction

The Palaeozoic sediments of Sweden constitute isolated remnants on the stable pre-Cambrian Baltic Shield. This paper is concerned with the Lower Ordovician, the so-called "Orthoceras Limestone", of the provinces of Närke, Östergötland and Öland (Fig. 1). Sampling spans the Arenig and Llanvirn epochs, and descriptions of these rocks have been given by Tjernvik (1956), Magnusson et al. (1962) and Jaanusson (1972). The limestones are rather pure, with 80–90% $CaCO_3$, formed by very slow sedimentation of carbonate mud or coarser skeletal fragments. Their colour is red, brownish-grey or grey, sometimes with green glauconitic spots. These flat-lying, virtually undisturbed limestones have been preserved due to slight downfaulting at the margins of the Palaeozoic areas in Närke and Östergötland, or due to the slight tilt of the pre-Cambrian basement in the Baltic Sea basin.

2. Method and results

The present palaeomagnetic study comprises 73 sites. Each site is represented by six samples (cores) drilled within an area of less than 0.1 m^2. Only one specimen was prepared from each core. The initial investigation of the limestones was undertaken in a "standard" way; i.e. thermal and alternating field (AF) demagnetization were performed on representative (or "pilot") specimens (As and Zijderveld, 1958; Creer, 1959; Irving et al., 1961) after the following plan:

thermal demagnetization at 100, 150, 200, 300, 400, 500, 550, 600 and, if suitable, 650°C

or:

AF demagnetization at 25, 50, 75, 100, 150, 200, 250, 300, (350), 400, 500, (600), (700) and 800 Oe peak field. Fields in brackets were only rarely used

The measurements were done using a Digico® spinner magnetometer (Molyneux, 1971). A few examples of this treatment are given in Fig. 2.

The next step in the investigation was to test the results for directional stability. This can be done either graphically, by plotting the directions in a suitable coordinate system or projection, or statistically, by calculating the most stable temperature or field range along with its stability index (Tarling and Symons, 1967). On the basis of these methods it was

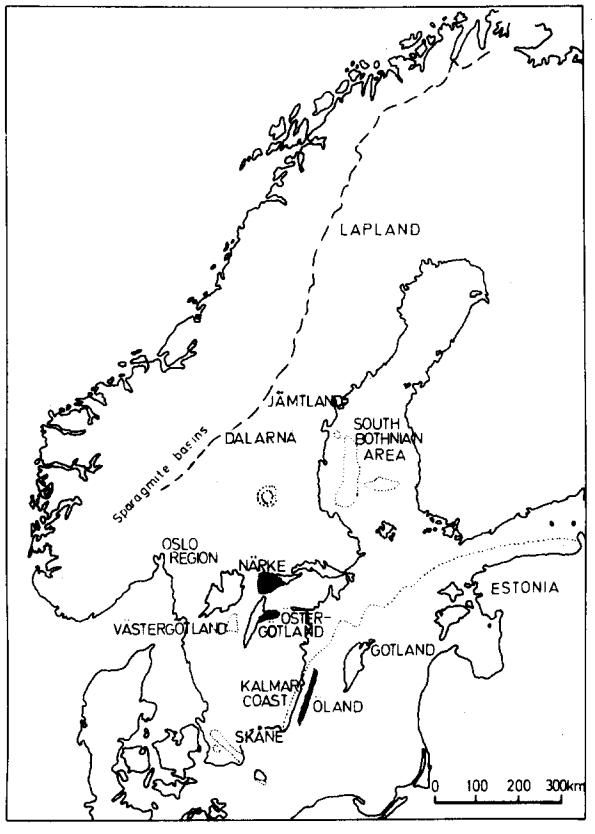


Fig. 1. The dark areas contain the Orthoceras Limestone treated in this paper. The Palaeozoic sedimentary areas in Sweden and the south Baltic Sea basin are enclosed by dotted lines. The dashed line delimits the Caledonian belt.

decided to demagnetize all specimens at 150 and 300°C (apart from the Närke rocks, which were treated at 150°C), or at 150 Oe.

The site mean directions following from these procedures are shown in Fig. 3A. The total mean direction and corresponding pole position calculated from these directions are:

$D = 132°$, $I = +70°$, $\alpha_{95} = 2.2°$, $N = 73$ sites, long. 46°E, lat. 30°N.

The α_{95} is the radius of the cone of confidence using a 95% confidence level (Fisher, 1953) and giving unit weight to sites.

Some features, however, cast doubt on the reliability of this pole position; for example, the smeared distribution and movement of the provincial mean directions (Fig. 3B) and the tendency to higher inclination (Fig. 3C) and higher apparent stability of direction for rocks treated by AF demagnetization (Fig. 2B). The only way to dispel any of these doubts was to investigate the stability spectrum in greater detail. Firstly, it was necessary to re-examine the results from the pilot specimens at lower temperatures than 400–500°C, see Fig. 4, to make sure that systematic directional movements had not been overlooked. This can easily happen, because the stability index is based on the three (or more) consecutive directions which show the closest grouping. Thus, a small, but systematic, movement is not distinguished from a movement distributed around the mean. A stable, non-composite magnetization should give small random motions (instrument error) around the mean at successive measurements until the remanence decays into the noise level. Consistent movements, however small, indicate that the remanence contains more than one component. For this reason, as Fig. 4 clearly shows, the above quoted mean direction *does not* represent a stable, single-component system. In the Öland specimens (Fig. 4C) this is not so obvious, because of the very slight movement and the overlapping confidence circles. This behaviour could be due to a similarity of the stability spectra of the various components and is not necessarily a sign of a stable one-component magnetization. The results have to be seen in the light of those from the other two provinces and the movements at temperatures above 500°C.

In order to analyse the components present we need also to examine the high-temperature part of the blocking temperature spectrum at close intervals (cf. Storetvedt, 1967). Accordingly, a second set of pilot specimens was chosen for detailed high-temperature demagnetization (no further AF demagnetization was undertaken), see examples shown in Fig. 5. Unfortunately, no single high-temperature component could be isolated, although a simple pattern reveals itself. The movements take place along two great circles: one NE–SW and another NNW–SSE. Both may be present in rocks from the same site (Fig. 5A) or even in the same specimen (Fig. 5D). In the Östergötland rocks, however, the NNW–SSE circle is dominant (Fig. 5B and C). It is important to stress that the prevailing trends and movements, often followed throughout the range of thermal demagnetization but being particularly pronounced at higher temperatures

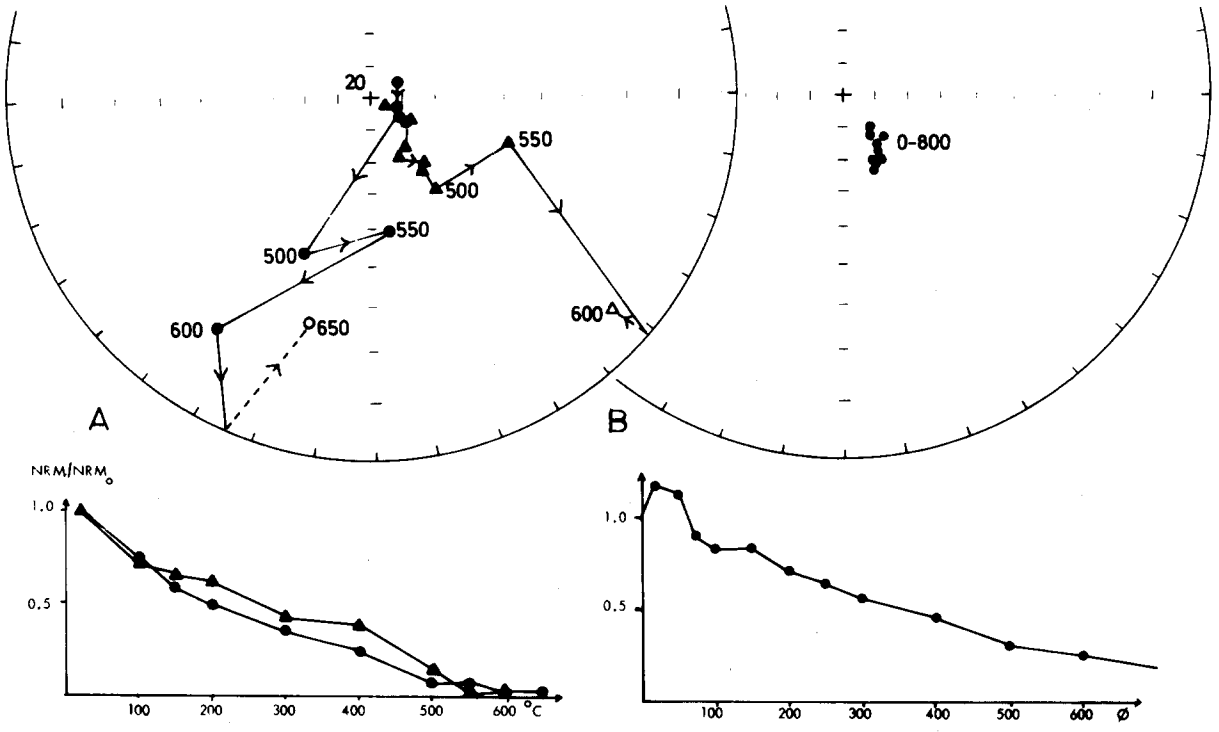


Fig. 2. Examples of directional and intensity changes during thermal and AF demagnetization of the Orthoceras Limestone. Stereographic projection with the circumference representing the horizontal. The full circles denote positive (down) inclination, the open negative (up).

A. Grey, calcilutitic limestone from Närke (•) and brownish-grey limestone from Östergötland (▲). "Stable" over the range 100–400°C, according to calculations after Tarling and Symons (1967). The initial NRM is $3.3 \cdot 10^{-6}$G and $3.7 \cdot 10^{-6}$G, respectively.

B. Grey, calcarenitic limestone from Öland. "Very stable" over the range 50–600 Oe. The initial NRM is $1.0 \cdot 10^{-6}$G.

($T > 550$°C), are well defined and cannot have been induced during the thermal treatment: the samples have been placed in the furnace in different positions to enable the detection of any induced component of magnetization that may be acquired in a small residual field. Increases in the magnetic viscosity often took place over 500°C (sometimes appearing at lower temperatures) giving rise to erratic intensity patterns, but these results have not been included in the analysis.

It is reasonable to assume that the NE–SW magnetization circle contains a magnetic axis with low inclination, because results from Cambrian and Silurian rocks of Norway (Storetvedt, 1967, 1973) and from two, three specimens of the Orthoceras Limestone in western Sweden (Claesson, in prep.) indicate the presence of a Lower Palaeozoic axis of magnetization pointing approximately 30°, −10°. The movement along the second circle is almost invariably to the SSE, but also in this case no stable end-point was reached. At the present stage it is difficult to speculate about its age and origin.

It has already been said that the AF-treated specimens show higher inclination and apparent stability. If an explanation of the dissimilar results from thermal and AF treatment were to be advanced in terms of chemical changes during heating, it must be said that the SSE and NE–SW movements have manifested themselves *before* any susceptibility increase occurring above 400°C. The AF demagnetization seems to be a less effective method to use for these limestones than thermal cleaning, as is also suggested by Biquand and Prévot (1971), Ouliac (1976) and Rao and Deutsch (1976). The thermal demagnetization, however, is difficult to apply to the desired extent owing to a very

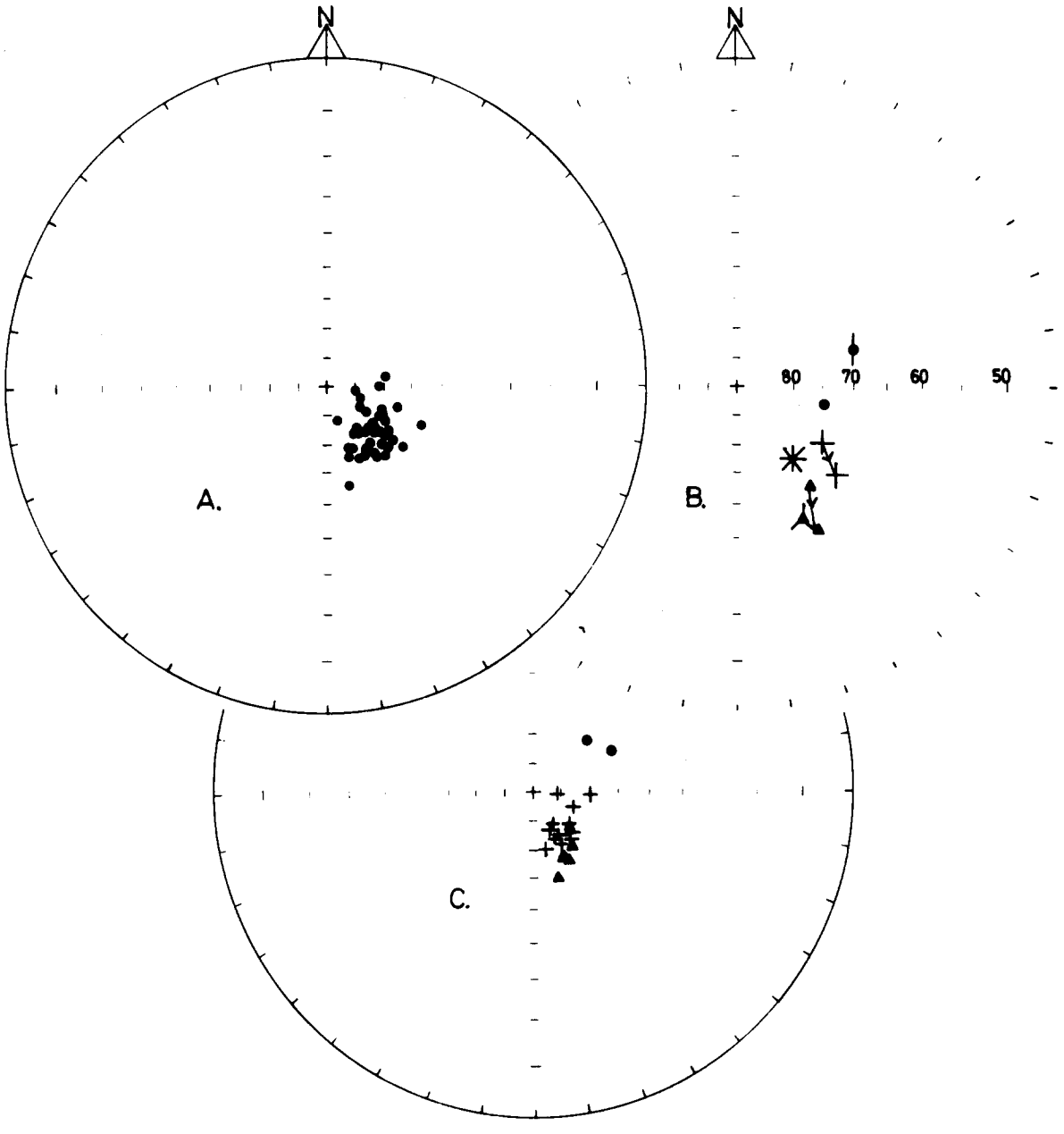


Fig. 3. Site mean or provincial mean (B) directions after bulk demagnetizations. All inclinations are positive.
A. After demagnetization at 150°C of five sites from Närke, at 300°C of ten sites from Östergötland and 31 sites from Öland.
B. Total mean direction for each province (note the change of scale); ⧫ = Närke, after AF demagnetization at 150 Oe of three sites; ● = Närke, after thermal demagnetization at 150°C of five sites; ⅄ = Östergötland, after AF demagnetization at 150 Oe of six sites; ▲ = Östergötland, after thermal demagnetization at 150 and 300°C of ten sites, the arrow indicates the movement from lower to higher temperature; * = Öland, after AF demagnetization at 150 Oe of thirteen sites; and + = Öland, after thermal demagnetization at 150 and 300°C of 31 sites, the arrow indicates the movement from lower to higher temperature.
C. After demagnetization at 150 Oe of three sites from Närke (●), of six sites from Östergötland (▲), and of thirteen sites from Öland (+).

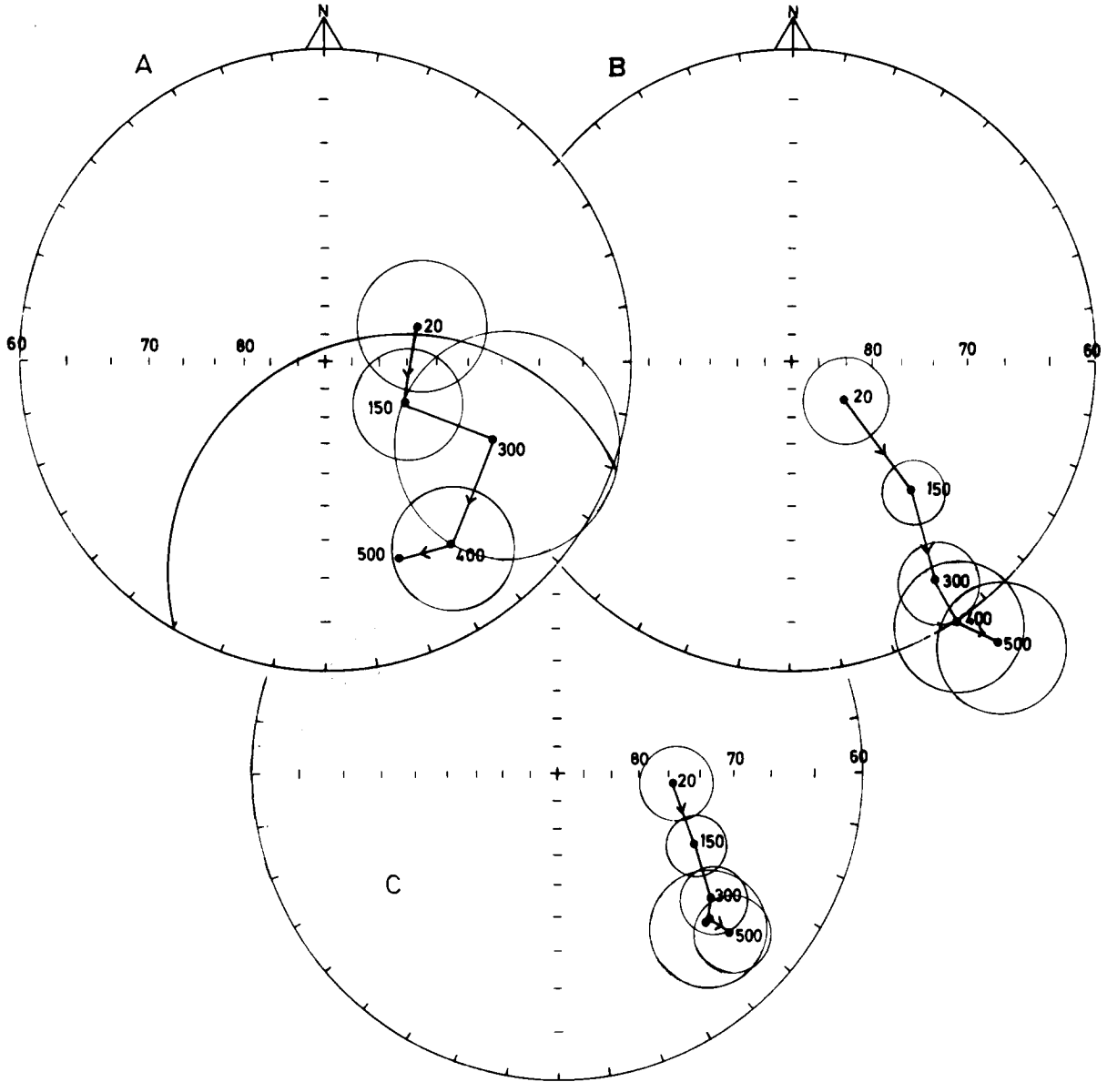
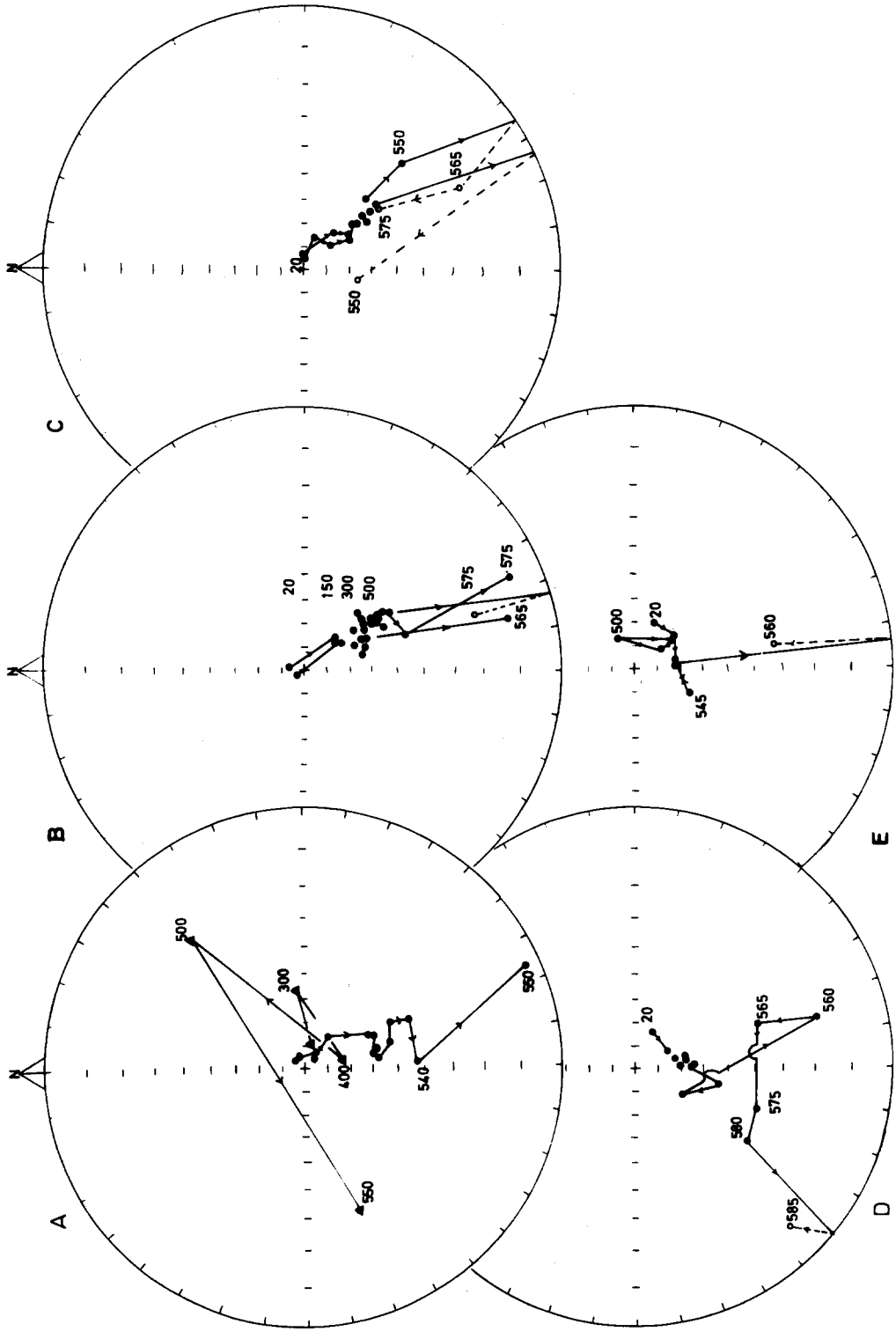


Fig. 4. Mean directions of all pilot specimens for each province at different demagnetization levels. All inclinations are positive (down). The circles denote the values of α_{95}. Note the change of scale.
A. Närke, nine specimens. The large value of α_{95} is due to the specimens moving apart in a NE–SW system.
B. Östergötland, eleven specimens.
C. Öland, 30 specimens.

low intensity of magnetization and the appearance of viscosity at higher temperatures. Possibly, a combination of the two methods may prove more efficient in separating the various components.

3. Conclusion

There are at least two possibilities as regards the interpretation of the steeply inclined, fairly stable, bulk


A
B
C
D
E
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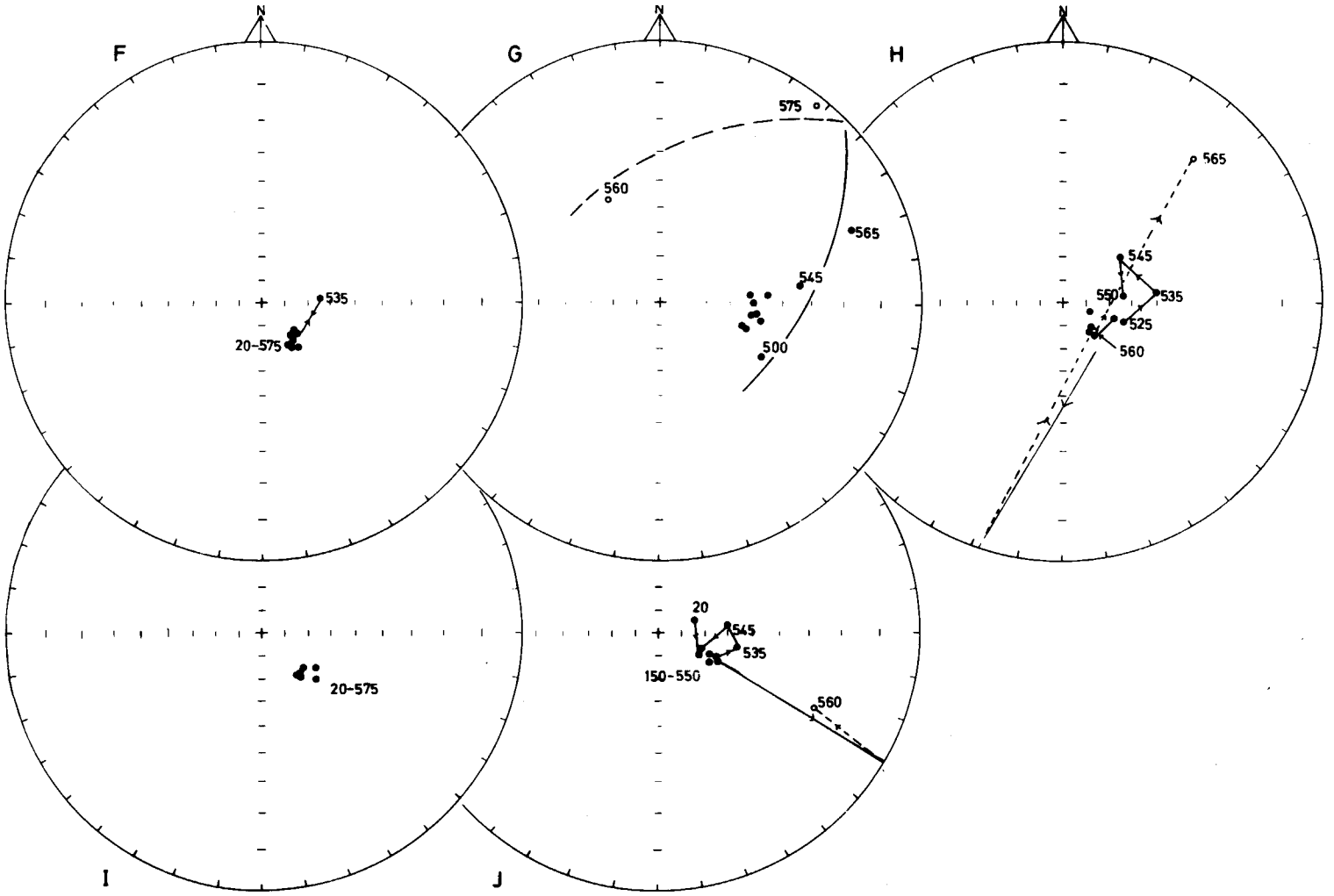


Fig. 5. Examples of directional changes during thermal demagnetization at more closely spaced intervals at higher temperatures than in Fig. 2. Denotations as in Fig. 2. The intensity decays are similar to those in Fig. 2A.

A. Närke, two specimens from the same locality.

B. and C. Östergötland.

D.–J. Öland; in (E) and (J), the final movement is not necessarily in the plane indicated. In (H) the direction perform a 180° change from 550 to 560°C.

direction of magnetization in the Orthoceras Limestone:

(1) It may dominantly constitute a multicomponent system formed by the normal and reverse subcomponents – with closely overlapping stability spectra – of one or more palaeomagnetic axes.

(2) It may dominantly consist of a steep palaeomagnetic component, together with a subordinate, multicomponent system as in (1).

From the available experimental evidence it appears impossible to chose between these two alternatives. However, whatever the explanation may be, it has been shown that great care should be taken to ensure that a truly one-component magnetization is present. A low value of α_{95} combined with AF treatment only may be a completely irrelevant guide in this respect.

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